

EE4340: Microgrids

Design Project



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1 Design a grid connected solar PV system to your home, assuming you are having a 3-phase supply

1.1 Size the solar panel based on the available Rooftop Area



Figure 1:Rooftop View

The Trina Solar 300 W monocrystalline module is selected for the rooftop installation. Key mechanical dimensions from the manufacturer's data sheet are:

- **Module dimensions:** 1658 mm × 992 mm × 6 mm
Converted to meters:

$$L = 1.658 \text{ m}, W = 0.992 \text{ m}$$

- **Module area:**

$$A_{\text{module}} = 1.658 \times 0.992 \text{ m}^2 = 1.645 \text{ m}^2$$

These dimensions are used to determine how many modules can be physically accommodated on the available rooftop surfaces.

The roof consists of four usable sections:

Roof Section	Dimensions (ft)	Dimensions (m)
Roof A (×2)	18 ft × 45 ft	5.486 m × 13.716 m
Roof B (×2)	14 ft × 9 ft	4.267 m × 2.743 m

Conversion factor used:

$$1 \text{ ft} = 0.3048 \text{ m}$$

Large Roof Sections: (18 ft × 45 ft)

- Roof size in meters:

$$\mathbf{5.486\ m \times 13.716\ m}$$

Panels are arranged in **portrait orientation**, placing the shorter side (0.992 m) across the width.

Number of panels across width:

$$N_{\text{width}} = \left(\frac{5.486}{0.992} \right) \approx 5.53 = 5$$

Number of panels along length:

$$N_{\text{length}} = \left(\frac{13.716}{1.658} \right) \approx 8.27 = 8$$

Panels per large roof section:

$$N_{\text{large}} = 5 \times 8 = 40$$

Since there are two such roof sections:

$$N_{\text{large,total}} = 2 \times 40 = \mathbf{80\ modules}$$

Small Roof Sections: (14 ft × 9 ft)

- Roof size in meters:

$$\mathbf{4.267\ m \times 2.743\ m}$$

Using the same portrait orientation:

Panels across width:

$$N_{\text{width}} = \left(\frac{2.743}{0.992} \right) \approx 2.76 = 2$$

Panels along length:

$$N_{\text{length}} = \left(\frac{4.267}{1.658} \right) \approx 2.57 = 2$$

Panels per small roof section:

$$N_{\text{small}} = 2 \times 2 = 4$$

Two such roof sections:

$$N_{\text{small,total}} = 2 \times 4 = 8 \text{ modules}$$

Total Installed PV Capacity

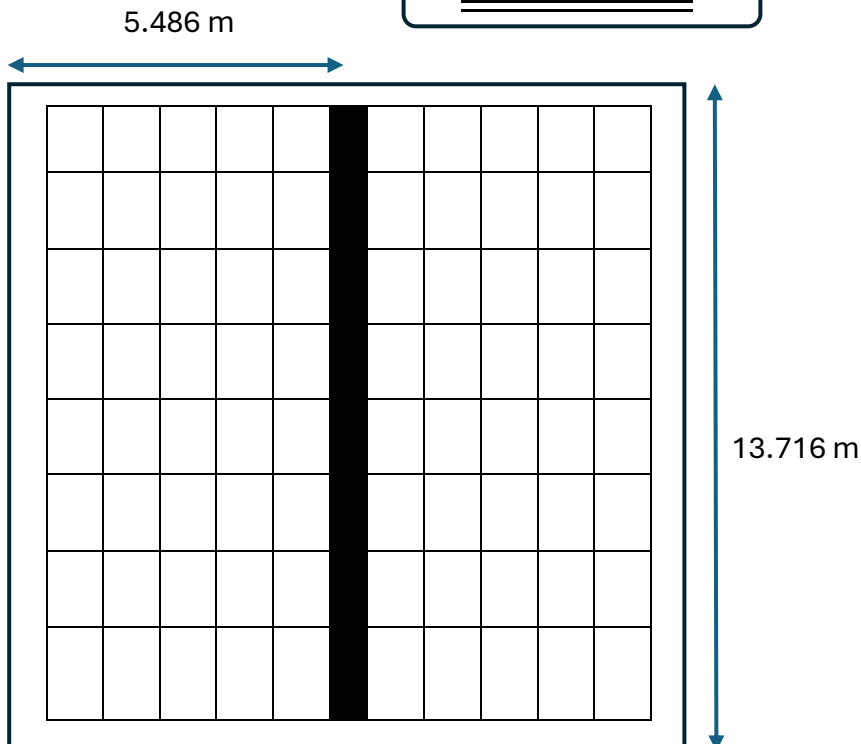
Total number of modules:

$$N_{\text{total}} = 80 + 8 = 88 \text{ modules}$$

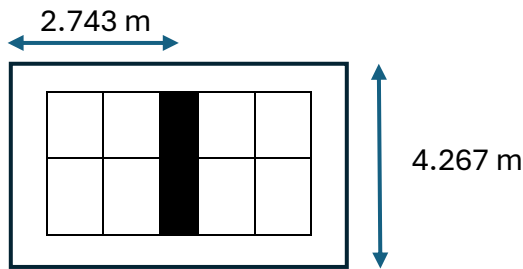
Given each module is **300 Wp**, the total system capacity is:

$$P_{\text{dc}} = 88 \times 300 \text{ W} = 26,400 \text{ W}$$

$$\underline{\underline{P_{\text{dc}} = 26.4 \text{ kWp}}}$$



**Panel arrangement in Large
Roof Sections**



Panel arrangement in Small Roof Sections

1.2 Sizing the Inverter

SUN2000-12/15/17/20/25KTL-M5 Technical Specification					
Technical Specification	SUN2000 -12KTL-M5	SUN2000 -15KTL-M5	SUN2000 -17KTL-M5	SUN2000 -20KTL-M5	SUN2000 -25KTL-M5
Efficiency					
Max. efficiency	98.4%	98.4%	98.4%	98.4%	98.4%
European weighted efficiency	97.9%	98.0%	98.1%	98.1%	98.2%
Input					
Recommended max. PV power ¹	18,000 Wp	22,500 Wp	25,500 Wp	30,000 Wp	37,500 Wp
Max. input voltage ²			1100 V		
Full-load MPPT voltage range	370V~800V	410V~800V	440V~800V	480V~800V	530~800V
MPPT Operating voltage range ³			200 V ~ 1000 V		
Start-up voltage			200 V		
Rated input voltage			600 V		
Max. input current per MPPT			30 A (two string) / 20 A (single string)		
Max. short-circuit current			40 A		
Number of MPP trackers			2		
Max. number of inputs			4		
Output					
Grid connection			Three phase		
Rated output power	12,000 W	15,000 W	17,000 W	20,000 W	25,000 W
Max. apparent power	13,200 W	16,500 VA	18,700 VA	22,000 VA	27,500 VA
Rated output voltage	220 Vac / 380 Vac, 230 Vac / 400 Vac, 239.6 Vac / 415Vac, 3W + N + PE				
Rated AC grid frequency	50 Hz / 60 Hz				
Max. output current	18.2A/380Vac 17.3A/400Vac 16.7A/415Vac	25.2A/380Vac 23.9A/400Vac 23.1A/415Vac	28.6A/380Vac 27.1A/400Vac 26.1A/415Vac	33.6A/380Vac 31.9A/400Vac 30.8A/415Vac	42.0A/380Vac 39.9A/400Vac 38.5A/415Vac
Adjustable power factor	0.8 leading ... 0.8 lagging				
Max. total harmonic distortion	≤ 3 %				

Figure 2: Inverter Data Sheet

In the case of the designed PV array, the overall installed DC power is 26.4 kWp (88 x 300 W modules). According to the technical information on the inverter, **Huawei SUN2000-20KTL-M5** is chosen as it can receive up to 30 kWp of DC input power, which is much more than the array capacity yet still offers a good price-performance ratio. Smaller models (12 kW, 15 kW, 17 kW) are dismissed since their recommended maximum PV power (18.0, 22.5 and 25.5 kWp respectively) is lower than the designed array power and thus would be overloaded on the DC side.

The 25 kW model, which is technically acceptable, is a \$0 of additional margin and an extra cost of equipment but with no proportionate increase in performance. Thus, the most suitable option is **SUN2000-20KTL-M5**, which is a safe, low-energy conversion, and well-utilized inverter throughout its operating range.

DC/AC Ratio Calculation

The DC/AC ratio is calculated as:

$$\text{DC/AC ratio} = \frac{P_{\text{DC,array}}}{P_{\text{AC,rated}}}$$

Where

- $P_{\text{DC,array}} = 26.4 \text{ kW}$ (total PV array size)
- $P_{\text{AC,rated}} = 20.0 \text{ kW}$ (rated AC output power of **SUN2000-20KTL-M5**)

$$\text{DC/AC ratio} = \frac{26.4}{20.0} = 1.32$$

Thus, the system has a **DC/AC ratio of 1.32**. This level of DC oversizing is within the usual design range (≈ 1.1 – 1.4) used in grid-connected PV systems to maximize annual energy yield, since the array rarely operates at nameplate power due to temperature effects, irradiance variations and other losses.

ELECTRICAL DATA (STC)								
Peak Power Watts- P_{MAX} (Wp)*	280	285	290	295	300	305	310	315
Power Output Tolerance- P_{MAX} (W)	0 ~ +5							
Maximum Power Voltage- V_{MPP} (V)	31.7	31.8	32.2	32.5	32.7	32.9	33.1	33.3
Maximum Power Current- I_{MPP} (A)	8.84	8.96	9.02	9.08	9.18	9.26	9.37	9.46
Open Circuit Voltage- V_{OC} (V)	38.4	38.5	38.7	39.4	39.8	40.0	40.2	40.4
Short Circuit Current- I_{SC} (A)	9.44	9.49	9.59	9.75	9.81	9.84	9.86	9.88
Module Efficiency η_{m} (%)	17.0	17.3	17.6	17.9	18.2	18.5	18.8	19.2

STC: Irradiance 1000W/m², Cell Temperature 25°C, Air Mass AM1.5.
 *Measuring tolerance: $\pm 3\%$.

Figure 3:Electrical Data of Solar Module at STC

TEMPERATURE RATINGS

NOCT(Nominal Operating Cell Temperature)	44°C (±2°C)
Temperature Coefficient of P _{MAX}	- 0.39%/°C
Temperature Coefficient of V _{oc}	- 0.29%/°C
Temperature Coefficient of I _{sc}	0.05%/°C

MAXIMUM RATINGS

Operational Temperature	-40~+85°C
Maximum System Voltage	1500V DC (IEC) 1000V DC (UL)
Max Series Fuse Rating	15A(Power ≤290W) 20A(Power ≥295W)

(DO NOT connect Fuse in Combiner Box with two or more strings in parallel connection)

Figure 4: Temperature Ratings and Maximum Ratings

Minimum number of modules per String:

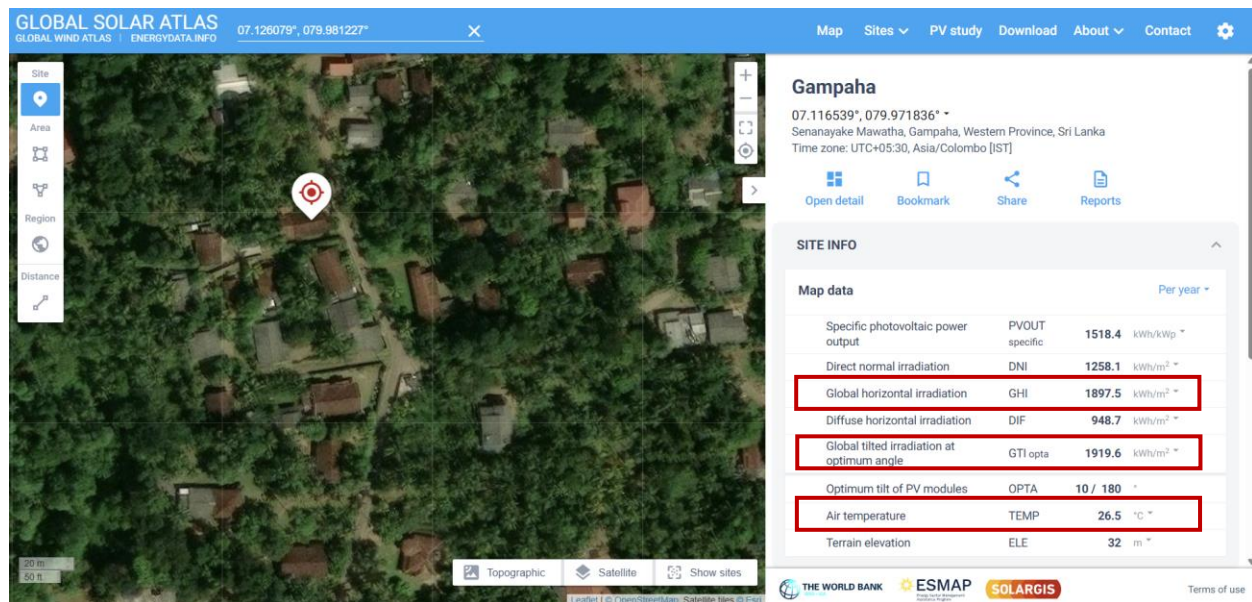


Figure 5: Data Obtained from Global Solar Atlas

Values of T_r

Installation of Array Frame	T_r
Ground Mounted Array	25°C
Array on roof where the array tilt angle is at least 20 degrees greater than the actual roof	25°C
Array structure parallel to roof with air gap greater than 150 mm	30°C
Array structure parallel to roof with air gap less than 150 mm	35°C

Figure 6: T_r Value Table

Assuming array structure is parallel to roof with air gap greater than 150mm;

$$\text{Max } T_{\text{Cell, effective}} = T_{a_max} + T_r = 50 + 30 = 80^\circ\text{C}$$

T_{a_max} is considered as 50°C giving a safety margin.

- MPP voltage temperature coefficient = $0.39\% / ^\circ\text{C}$ (V_{mp})

(Voltage temperature coefficient is not available, So P_{mp} temperature coefficient is used)

- $\% \Delta V_{\text{mp}} = 0.39 \times (T_{\text{eff}} - T_{\text{STC}}) = 0.39 \times (80 - 25) = 21.45\%$
- $\Delta V_{\text{mp}} = 32.7 \times 21.45\% = 7.01415 \text{ V}$
- $V_{\text{mp, eff of the module due to temperature}} = V_{\text{mp}} - \Delta V_{\text{mp}} = (32.7 - 7.01415) \text{ V} = 25.68585 \text{ V}$

Voltage drop in DC cabling between the array and inverter is assumed negligible.

Therefore, actual voltage at the input of the inverter = 25.68585 V

$$\text{Minimum no of modules in a string} = \frac{\text{effective minimum operating voltage of the MPPT}}{\text{effective minimum MPP voltage input of a module at the inverter}}$$

$$\text{Minimum no of modules in a string} = \frac{200+20 \text{ (safety margin of 10\%)}}{25.68585} = 8.565$$

➤ Therefore, minimum number of modules per string should be 9.

Maximum number of modules per String:

$$\text{Min } T_{\text{Cell, effective}} = T_{a_max} + T_r = -10 + 30 = 20^\circ\text{C}$$

T_{a_min} is considered as -10°C giving a safety margin.

Minimum effective cell temperature = 20°C

V_{oc} temperature coefficient = $0.29\% / ^\circ\text{C}$

$$\% \Delta V_{\text{oc}} = 0.29 \times (T_{\text{eff}} - T_{\text{STC}}) = 0.29 \times (20 - 25) = -1.45\%$$

$$\Delta V_{\text{oc}} = -39.8 \times 1.45\% = -0.5771 \text{ V}$$

$$V_{\text{oc}} \text{ (effective) of the module due to Temperature} = V_{\text{oc}} - \Delta V_{\text{oc}} = 39.8 + 0.5771 = 40.3771 \text{ V}$$

$$\text{maximum number of modules in a string} = \frac{\text{maximum operating voltage of the inverter}}{\text{effective maximum open circuit voltage for each module}}$$

$$\text{maximum number of modules in a string} = \frac{1100}{40.3771} = 27.243$$

➤ Therefore, maximum number of modules in string should be 27.

From the voltage and temperature limits of the module and inverter, the allowable number of modules in one series string was found to be:

$$N_{\min} = 9, N_{\max} = 27$$

So, any string length between 9 and 27 modules is acceptable.

The total number of modules in the system is:

$$N_{\text{total}} = 88 \text{ modules}$$

To use the inverter inputs efficiently, the array is divided into **4 equal strings**.

Number of modules per string:

$$N_{\text{string}} = \frac{N_{\text{total}}}{4} = \frac{88}{4} = \mathbf{22 \text{ modules/string}}$$

This satisfies the allowed range:

$$9 \leq N_{\text{string}} \leq 27 \text{ (string length is acceptable)}$$

The MPP current of one module is about $I_{\text{mpp,mod}} \approx 9 \text{ A}$.

In a series string the current is the same, so:

$$I_{\text{mpp,string}} \approx 9 \text{ A}$$

Each MPPT has two strings in parallel, so per-MPPT current is roughly:

$$I_{\text{MPPT}} \approx 2 \times 9 = \mathbf{18 \text{ A}}$$

which is below the inverter's maximum input current limit (30 A for two strings), so the current is also acceptable.

The array is finally arranged as:

- **4 strings × 22 modules in series per string**
- Each string power:

$$P_{\text{string}} = 22 \times 300 \text{ W} = 6.6 \text{ kW}$$

- Total array power:

$$P_{\text{array}} = 4 \times 6.6 = 26.4 \text{ kWp}$$

The inverter has **two MPPT inputs**, each accepting two strings. The strings are connected as:

- **MPPT 1:** String 1 // String 2 (two parallel strings of 22 modules)
- **MPPT 2:** String 3 // String 4 (two parallel strings of 22 modules)

1.3 Calculation of the energy generated per month from the roof-top solar PV system and how much energy we have to purchase from the grid or how much energy we can feed into the grid for a selected month

Energy generated per hour from the Roof-Top Solar PV System can be calculated using the following formula.

$$E_{sys} = \left(\frac{G_T}{G_{STC}} \right) \times P_{array,STC} \times f_{temp} \times f_{man} \times f_{dirt} \times f_{inv} \times f_{wire}$$

Where:

E_{sys} = average yearly energy output of the PV array, [Wh]

G_T = average solar radiation incident on the PV array [kWh/m²] for the selected site (allowing for tilt, orientation and shading)

G_{STC} = the incident radiation at standard test conditions [1 kW/m²]

P_{array_STC} = rated output power of the array under standard test conditions [W]

f_{temp} = temperature de-rating factor

f_{dirt} = de-rating factor for dirt

f_{man} = de-rating factor for manufacturing tolerance

$f_{mismatch, wire loss}$ = derating factor for the subsystem

f_{inv} = efficiency of the inverter

Calculation of derating factors:

•Temperature Derating Factor

The average effective daytime cell temperature of the solar module is calculated using the following method:

$$T_{cell,eff} = T_{a,day} + T_r$$

$$T_{cell,eff} = 26.5 + 30$$

$$T_{cell,eff} = 56.5 \text{ }^{\circ}\text{C}$$

Where,

$T_{cell,eff}$ - Average daytime effective cell temperature in Celsius.

$T_{a,day}$ - Daytime average ambient temperature for the month (Found using Global Solar Atlas)

T_r - Rise in temperature of the array when exposed to the sunlight.

$$\%T_{loss,temp} = Y(T_{cell,eff} - T_{STC})$$

$$\%T_{loss,temp} = 0.39(56.5 - 25)$$

$$\%T_{loss,temp} = 12.285\%$$

$$\therefore \text{Temperature derating factor } (f_{temp}) = 1 - T_{loss,temp}$$

$$f_{temp} = 1 - 0.1229$$

$$f_{temp} = 0.8771$$

•Manufacturer's output tolerance

In the given data sheet manufacturing tolerance is specified as $\pm 3\%$.

$$\therefore \text{Manufacturer derating factor } (f_{man}) = 1 - 0.03$$

$$f_{man} = 0.97$$

•Derating factor due to dirt

An acceptable value for power loss ($\%P_{loss,Dirt}$) would be 5% and can be used when there is no specific evidence of significant dirt or salt accumulation

$$f_{dirt} = 0.95$$

•Derating factor due to wiring losses and mismatch

For this calculation, the following value has been assumed for wiring losses and mismatch derating:

$$f_{wire} = 0.99$$

•Derating factor due to inverter losses

According to the manufacturer datasheet of the inverter, Huawei SUN2000-20KTL-M5 inverter has 98.4% maximum efficiency.

$$\text{Inverter derating factor } (f_{inv}) = 0.984$$

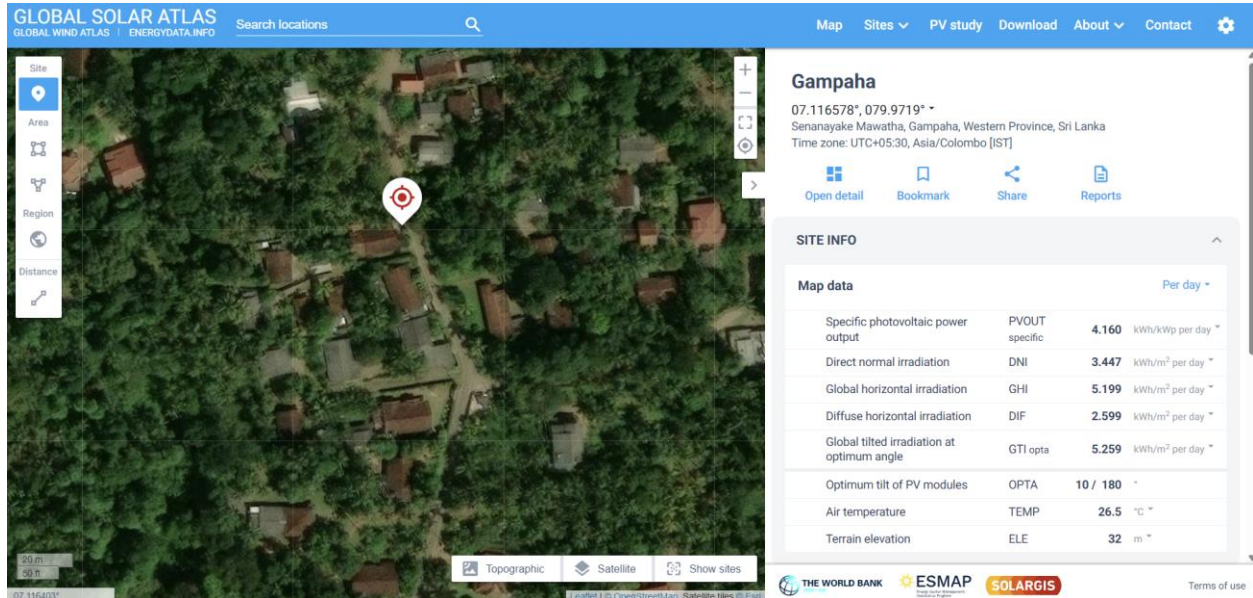


Figure 7: Global Solar Atlas Data

From Global Solar Atlas G_T was found to be 5.259 kWh/m² per day. (Assuming the panels are tilted at optimum angle)

The daily energy output from a single solar module is calculated using the following expression:

$$E_{\text{module,day}} = \frac{5.259}{1} \times 300 \times 0.8771 \times 0.97 \times 0.95 \times 0.984 \times 0.99$$

$$E_{\text{module,day}} = 1242.22 \text{ Wh}$$

The daily energy output of the entire solar PV system can be calculated using the following expression:

$$E_{\text{system,day}} = E_{\text{module,day}} \times 88$$

$$E_{\text{system,day}} = 1256.064 \times 88 = 109.315 \text{ kWh}$$

Energy generated by the solar PV system during a single month can be calculated as follows.

$$E_{\text{system,month}} = \frac{E_{\text{system,day}} \times 365}{12}$$

$$\underline{\underline{E_{\text{system,month}} = 3324.99 \text{ kWh} \approx 3325 \text{ kWh}}}$$

Based on the electricity consumption recorded over the past 10 months, the average household consumption per month has been calculated.

Average Electricity consumption of the house per month

$$= \frac{173 + 186 + 158 + 147 + 187 + 169 + 147 + 157 + 141 + 133}{10} \text{ kWh} = 159.8 \text{ kWh}$$

Since the proposed Solar PV system generates more energy than the household consumes, the surplus can be exported to the grid.

Excess energy that can be fed to the grid = $3325 - 159.8 = 3165.2 \text{ kWh}$

So, the proposed rooftop solar PV system can export approximately **3165 kWh** of energy to the grid each month.

2 Design verification using PVSyst Software

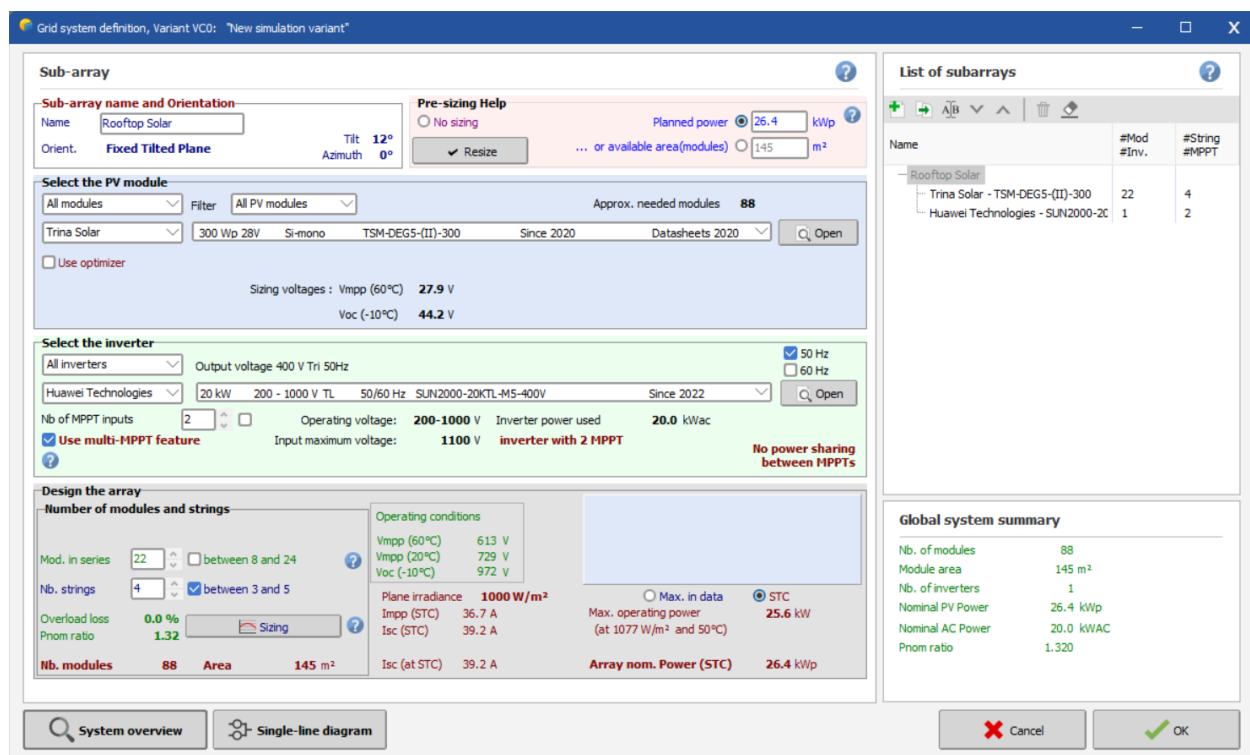
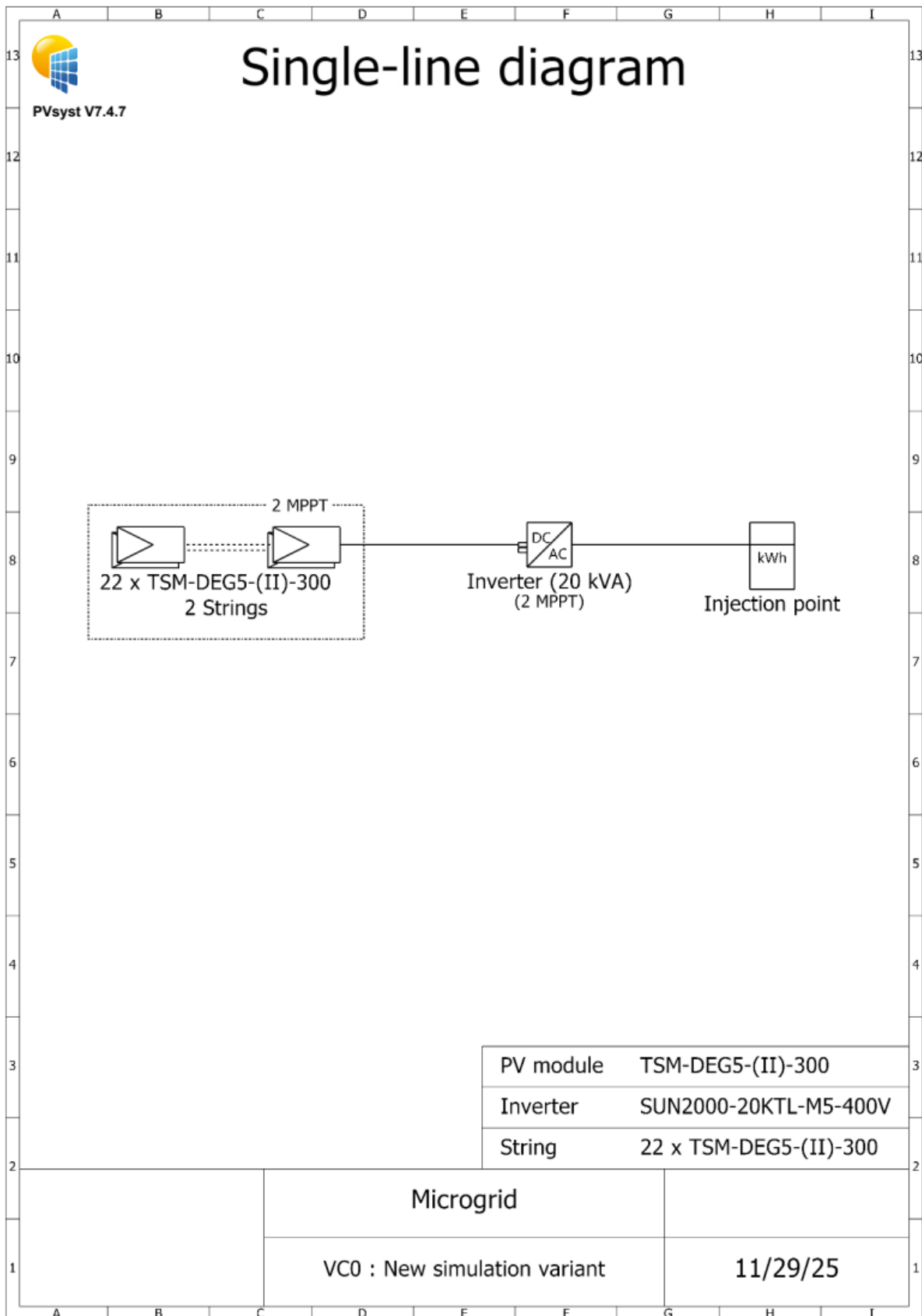


Figure 8: System Design in PVSyst Software



Report Generated from PVSyst Software



Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Microgrid Design Project

Department of Electrical Engineering

No 3D scene defined, no shadings

System power: 26.40 kWp

Walpola - Sri Lanka

| K.T.S.Kaushalya 210280T

**PVsyst V7.4.7**

VC0, Simulation date:
11/29/25 14:12
with V7.4.7

Project: Microgrid

Variant: New simulation variant

Project summary**Geographical Site**

Walpola

Sri Lanka

Situation

Latitude 7.12 °N

Longitude 79.97 °E

Altitude 20 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

Asgiriwalpola

Meteonorm 8.1 - Synthetic

System summary**Grid-Connected System**

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane

Tilt/Azimuth 12 / 0 °

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

System information**PV Array**

Nb. of modules

88 units

Pnom total

26.40 kWp

Inverters

Nb. of units

1 unit

Pnom total

20.00 kWac

Pnom ratio

1.320

Results summary

Produced Energy 42107 kWh/year Specific production 1595 kWh/kWp/year Perf. Ratio PR 82.72 %

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Single-line diagram	12



PVsyst V7.4.7
VC0, Simulation date:
11/29/25 14:12
with V7.4.7

Project: Microgrid
Variant: New simulation variant

General parameters

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Orientation

Fixed plane

Tilt/Azimuth 12 / 0 °

Sheds configuration

No 3D scene defined

Models used

Transposition Perez

Diffuse Perez, Meteonorm

Circumsolar separate

Horizon

Free Horizon

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer

Trina Solar

Model

TSM-DEG5-(II)-300

(Original PVsyst database)

Unit Nom. Power

300 Wp

Number of PV modules

88 units

Nominal (STC)

26.40 kWp

Modules

4 string x 22 In series

At operating cond. (50°C)

Pmpp

23.83 kWp

U mpp

642 V

I mpp

37 A

Total PV power

Nominal (STC)

26 kWp

Total

88 modules

Module area

145 m²

Cell area

128 m²

Inverter

Manufacturer

Huawei Technologies

Model

SUN2000-20KTL-M5-400V

(Original PVsyst database)

Unit Nom. Power

20.0 kWac

Number of inverters

2 * MPPT 50% 1 unit

Total power

20.0 kWac

Operating voltage

200-1000 V

Max. power (=>48°C)

22.0 kWac

Pnom ratio (DC:AC)

1.32

No power sharing between MPPTs

Total inverter power

Total power

20 kWac

Number of inverters

1 unit

Pnom ratio

1.32

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)

20.0 W/m²K

Uv (wind)

0.0 W/m²K/m/s

DC wiring losses

Global array res.

290 mΩ

Loss Fraction

1.5 % at STC

Module Quality Loss

Loss Fraction

-0.8 %

Module mismatch losses

Loss Fraction

2.0 % at MPP

Strings Mismatch loss

Loss Fraction

0.1 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



PVsyst V7.4.7
VC0, Simulation date:
11/29/25 14:12
with V7.4.7

Project: Microgrid
Variant: New simulation variant

Main results

System Production

Produced Energy

42107 kWh/year

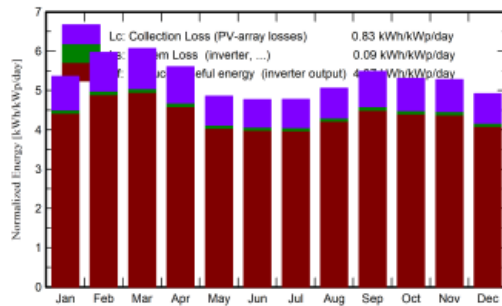
Specific production

1595 kWh/kWp/year

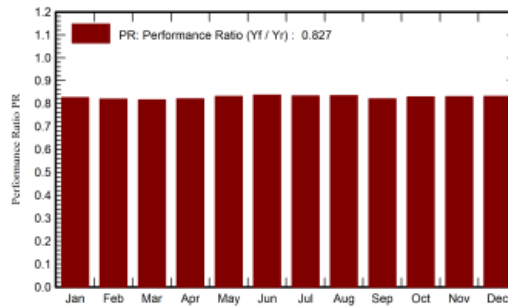
Perf. Ratio PR

82.72 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_Grid kWh	PR ratio
January	152.6	70.18	26.46	166.1	162.4	3693	3622	0.826
February	157.8	69.64	27.21	167.2	163.7	3690	3620	0.820
March	185.2	81.08	28.01	188.2	184.3	4137	4057	0.817
April	172.9	76.80	27.77	168.1	164.3	3713	3640	0.820
May	159.5	90.33	28.63	150.6	146.3	3373	3307	0.832
June	153.7	88.19	27.77	143.0	138.9	3226	3162	0.837
July	158.5	81.59	28.10	148.1	144.0	3324	3258	0.834
August	163.0	92.18	27.80	156.7	152.8	3523	3454	0.835
September	164.7	66.53	27.17	164.5	161.0	3636	3564	0.821
October	158.8	82.61	27.14	164.7	161.3	3676	3604	0.829
November	147.8	70.78	26.21	158.4	155.3	3540	3471	0.830
December	140.2	70.73	26.48	152.5	149.4	3414	3347	0.831
Year	1914.7	940.64	27.40	1928.1	1883.7	42944	42107	0.827

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

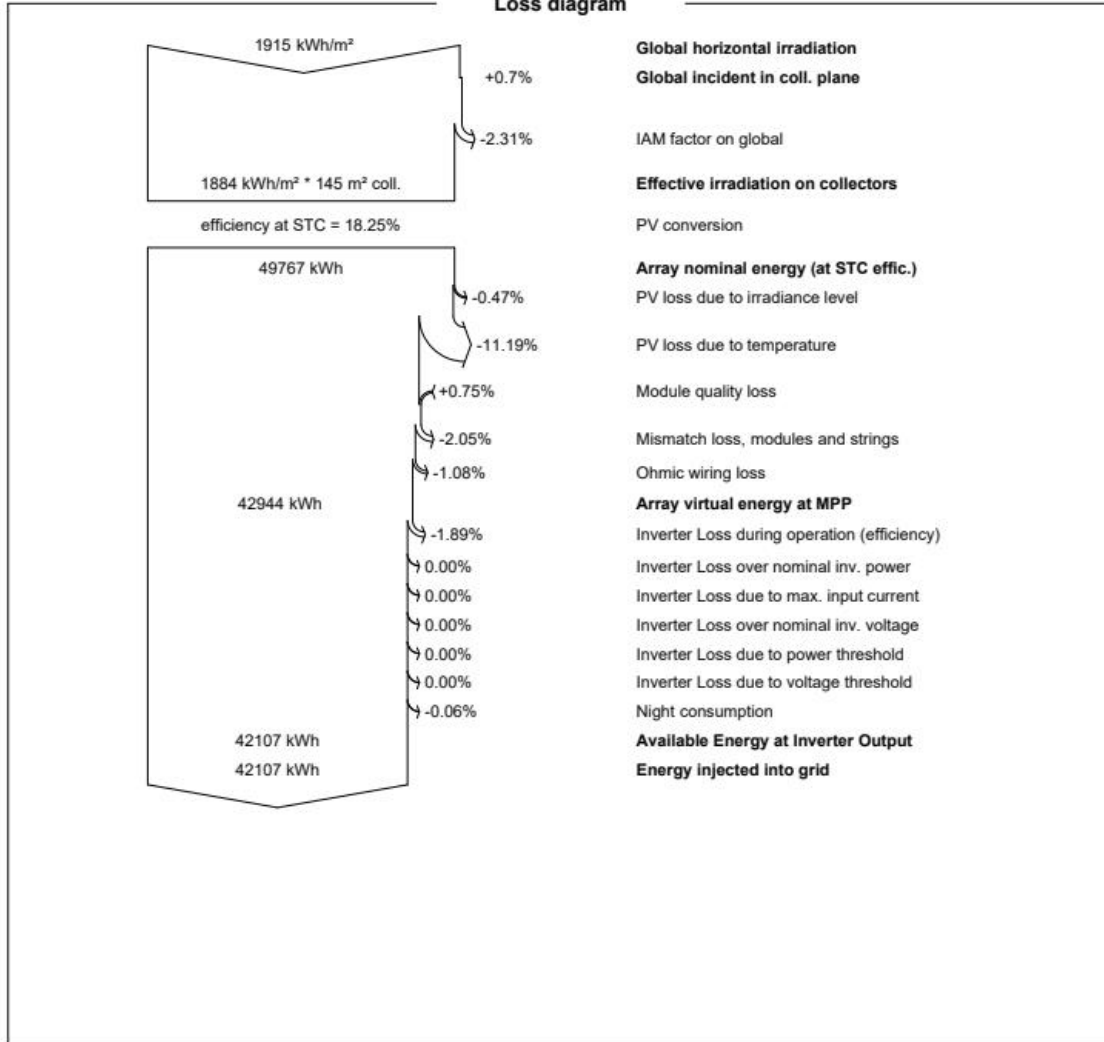
PR Performance Ratio



PVsyst V7.4.7
VC0, Simulation date:
11/29/25 14:12
with V7.4.7

Project: Microgrid
Variant: New simulation variant

Loss diagram





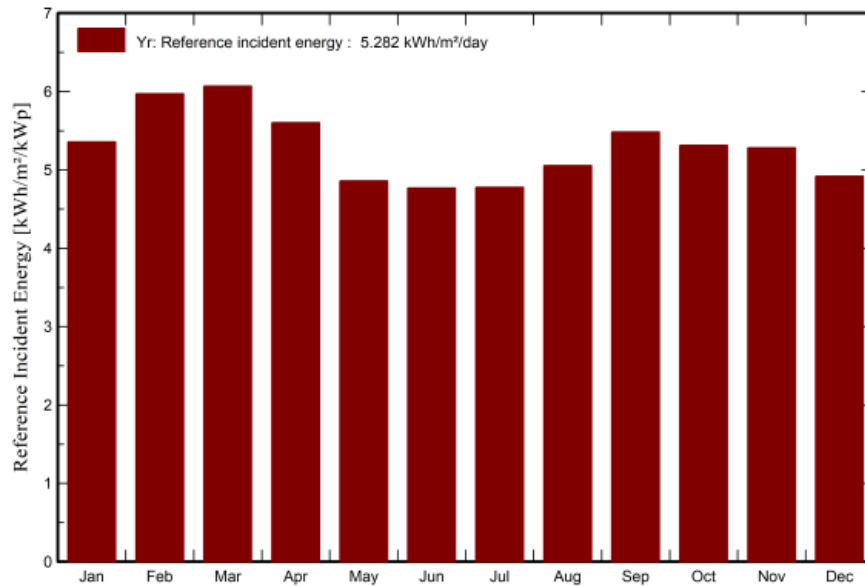
PVsyst V7.4.7
VC0, Simulation date:
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with V7.4.7

Project: Microgrid

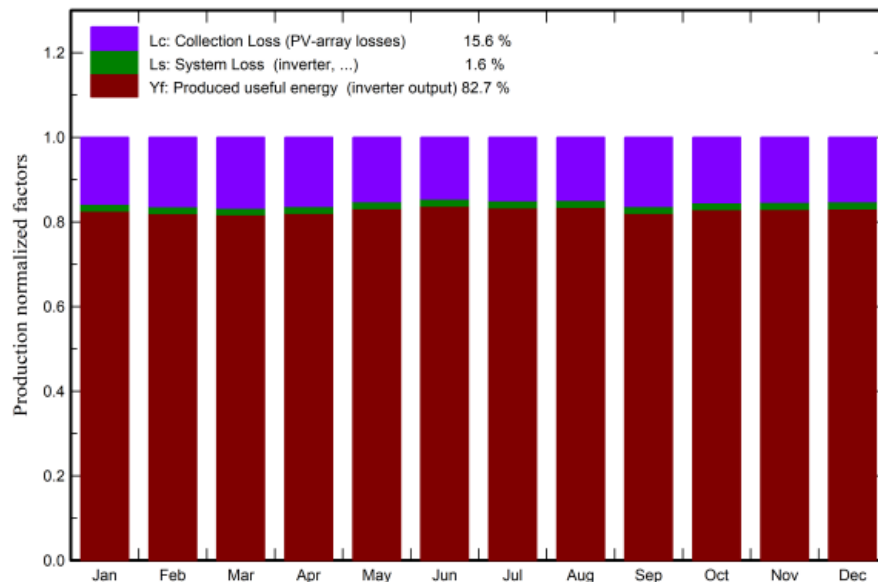
Variant: New simulation variant

Predef. graphs

Reference Incident Energy in Collector Plane



Normalized Production and Loss Factors



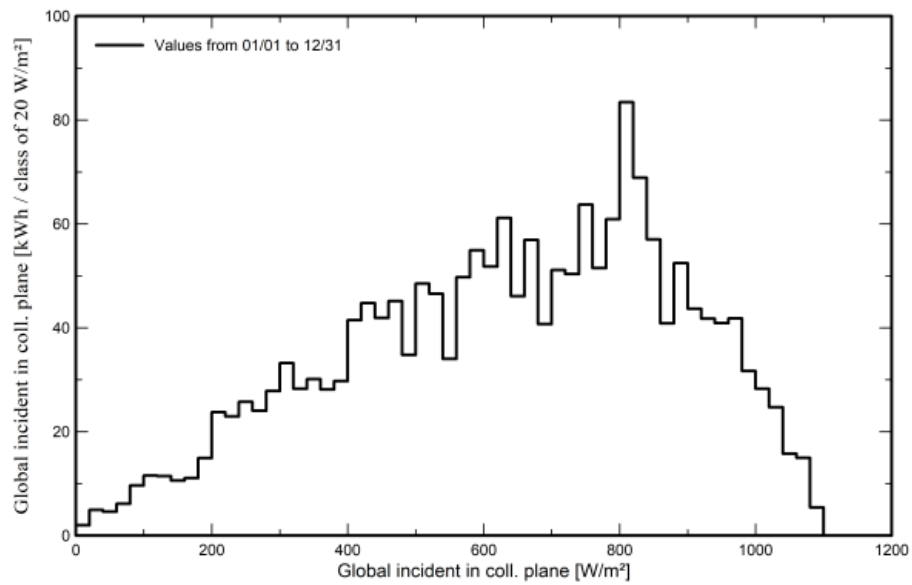


PVsyst V7.4.7
VC0, Simulation date:
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with V7.4.7

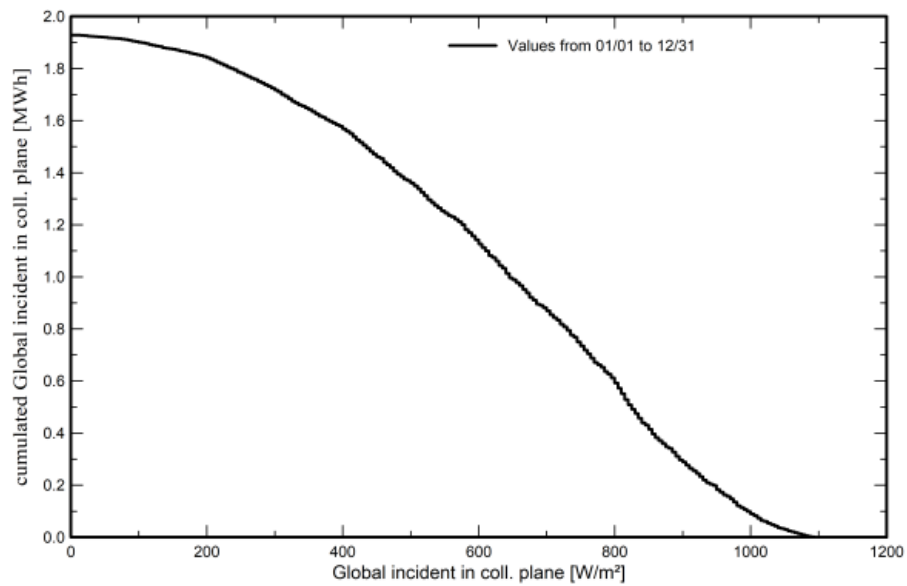
Project: Microgrid
Variant: New simulation variant

Predef. graphs

Incident Irradiation Distribution



Incident Irradiation cumulative distribution





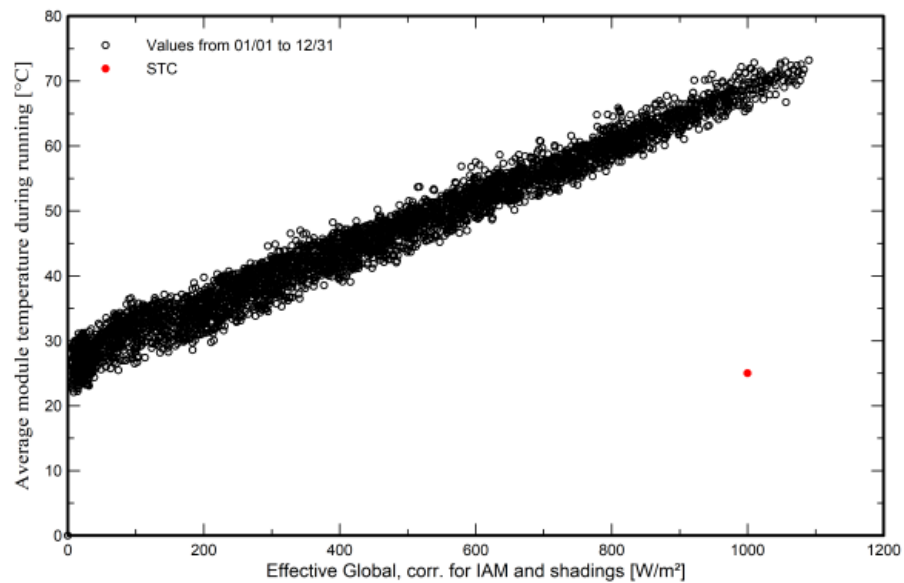
PVsyst V7.4.7
VC0, Simulation date:
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with V7.4.7

Project: Microgrid

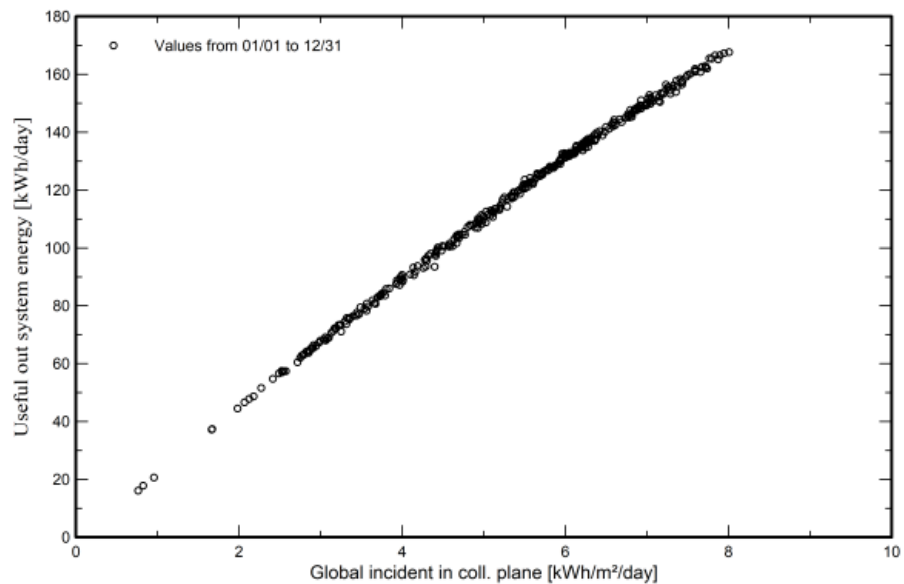
Variant: New simulation variant

Predef. graphs

Array Temperature vs. Effective Irradiance



Daily Input/Output diagram





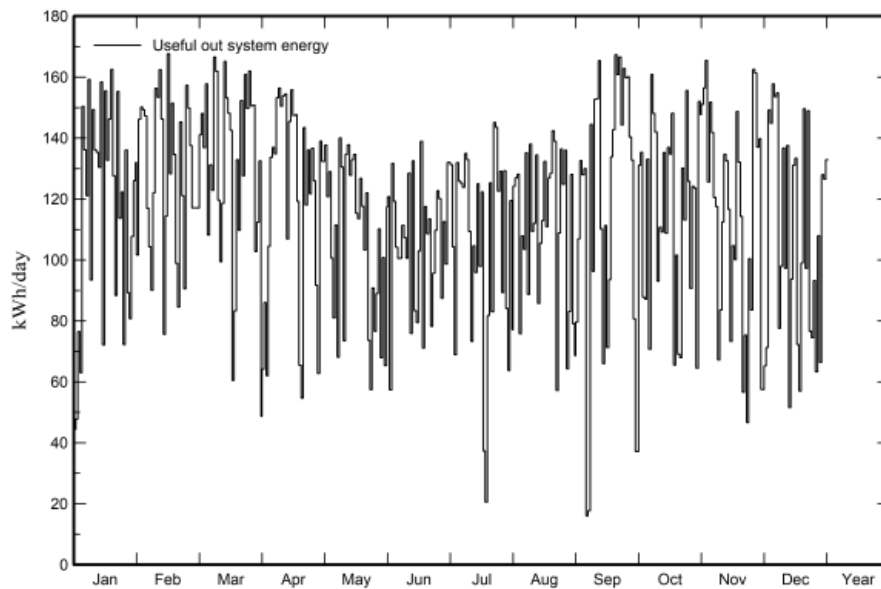
PVsyst V7.4.7
VC0, Simulation date:
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with V7.4.7

Project: Microgrid

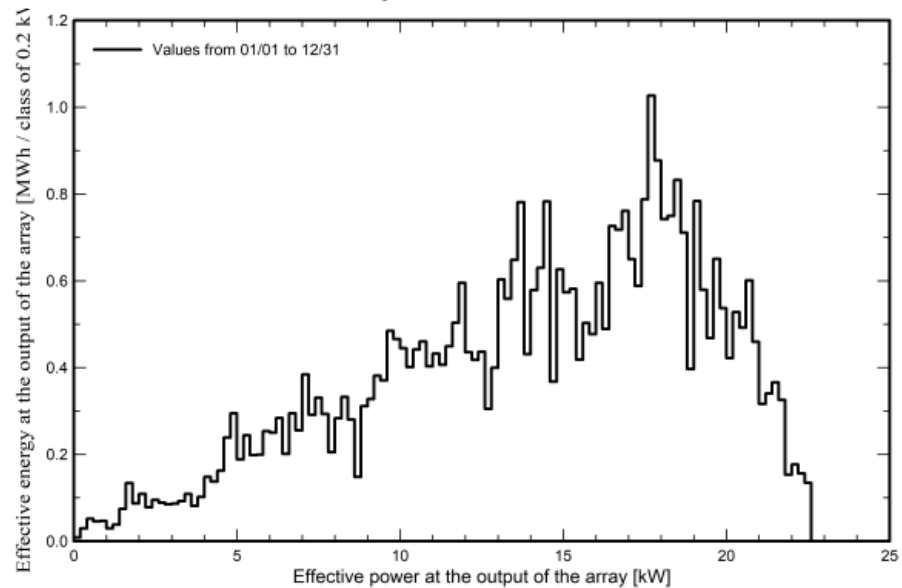
Variant: New simulation variant

Predef. graphs

Daily System Output Energy



Array Power Distribution





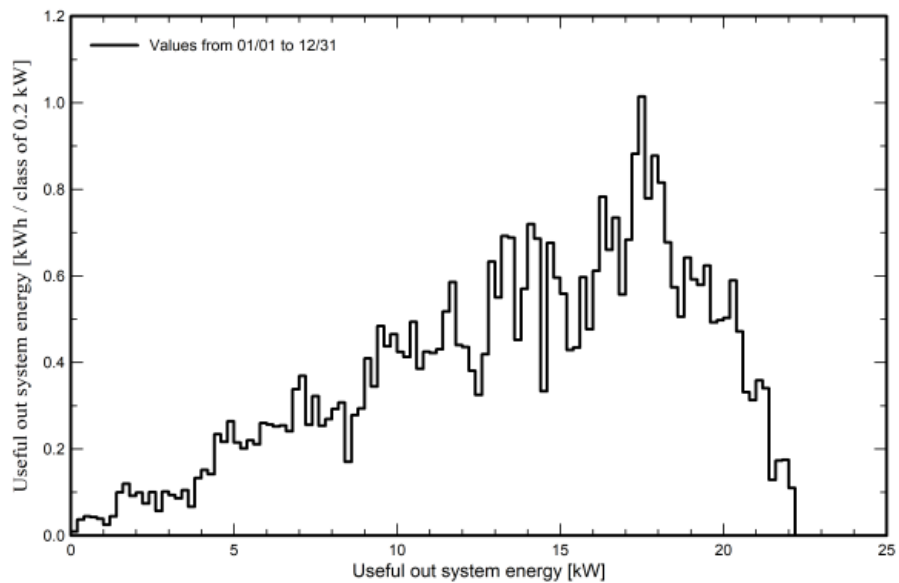
PVsyst V7.4.7
VC0, Simulation date:
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with V7.4.7

Project: Microgrid

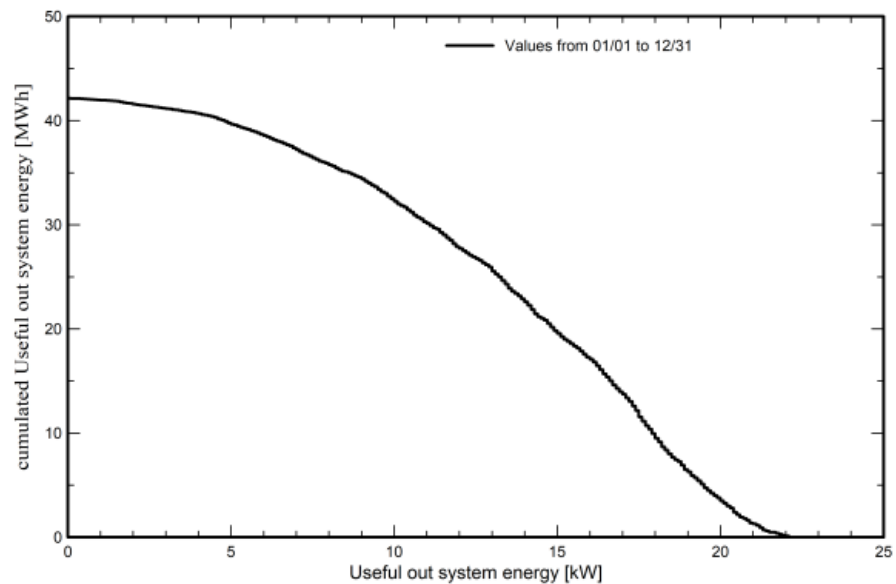
Variant: New simulation variant

Predef. graphs

System Output Power Distribution



System Output Power cumulative distribution



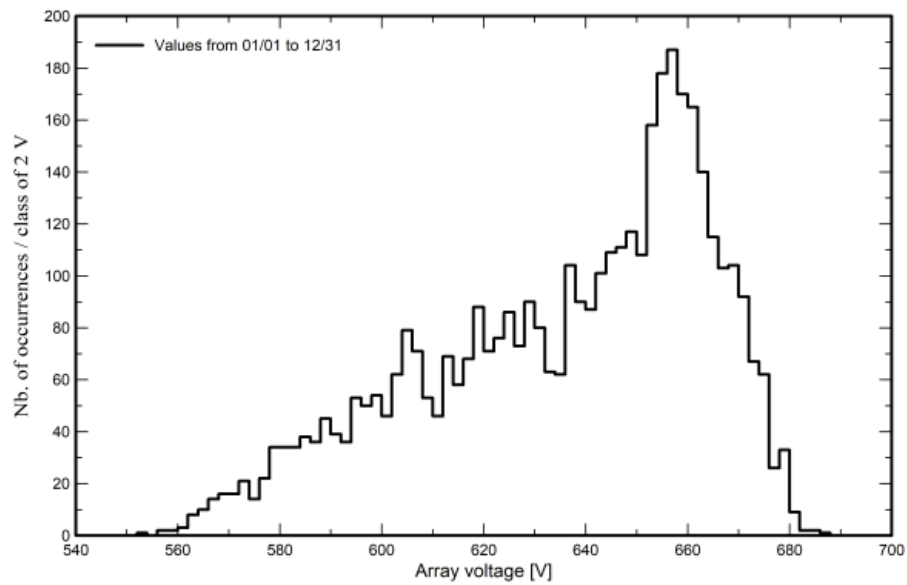


PVsyst V7.4.7
VC0, Simulation date:
11/29/25 14:12
with V7.4.7

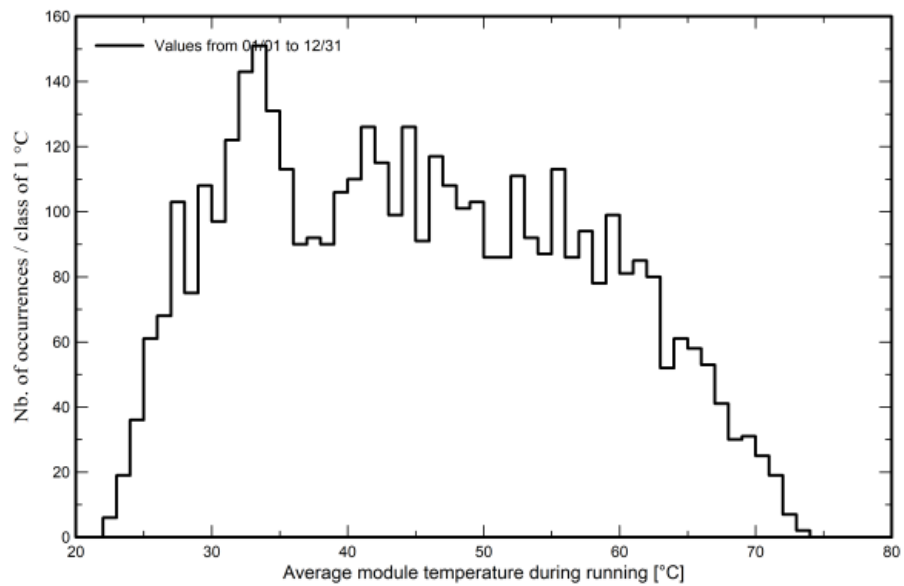
Project: Microgrid
Variant: New simulation variant

Predef. graphs

Array Voltage Distribution



Array Temperature Distribution during running



End of the PVSyst Report

2.1 Comparison between Calculated values and Results obtained from PVsyst Software

The manual energy analysis done on the 26.4 kWp PV system with derating factors including losses through temperature, wiring, inverter efficiency effects, and the effects of dirt led to the estimate of the monthly energy output of about 3325 kWh, which translates to the yearly energy output of about 39,900 kWh/year. Comparatively, the PVsyst simulation forecasts an energy production of 42,107 kWh/year annually and this is equivalent to an average production of approximately 3508 kWh/month. This means that the PVsyst estimation is approximately 5-6 % greater than the one estimated manually. Such discrepancy can be attributed to the fact PVsyst is a more comprehensive hour-by-hour model, with location-specific irradiance and module temperature models, inverter characteristics, and optical losses. It also considers seasonal changes in irradiance and temperature which are usually simplified with average values in manual calculation. All in all, the fact that the two results are close by supports that the manual estimation is viable and the PV system is likely to yield between 3300-3500 kWh per month in normal operating conditions.

3 Modeling the designed Roof-Top Solar PV system as a grid feeding source using PSCAD/ EMTDC

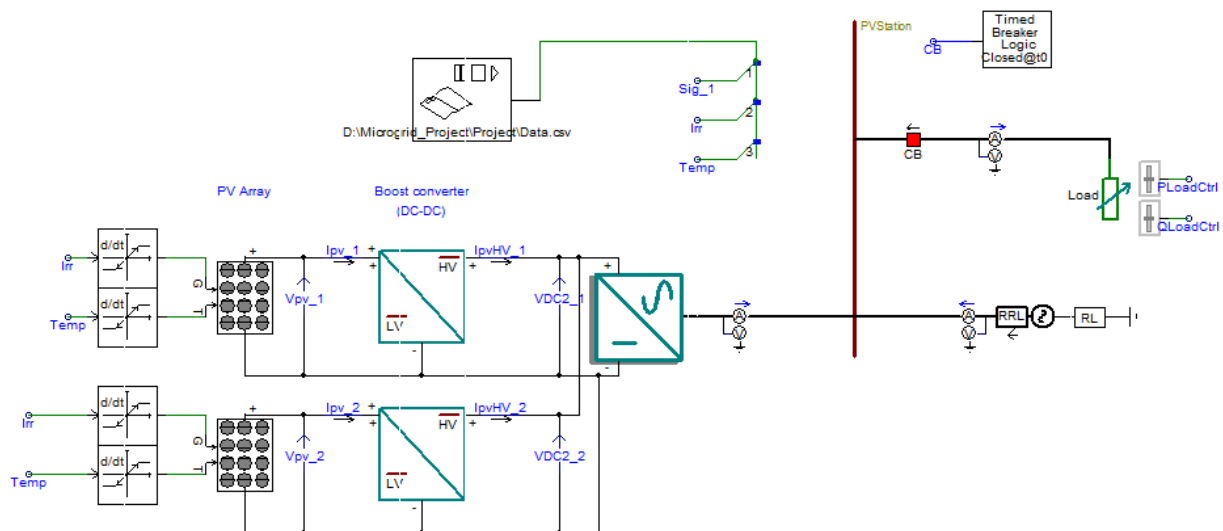


Figure 9:Single Line Diagram in PSCAD

PV array parameters

General

PV array name (optional)	PVarray
Number of modules connected in series per array	22
Number of module strings in parallel per array	2
Number of cells connected in series per module	60
Number of cell strings in parallel per module	1
Reference irradiation	1000
Reference cell temperature	25
Graphics Display	industry

Figure 10:PV Array Design

Main Data

General

PV Array Short Circuit Current	0.01962 [kA]
PV Array Open Circuit Voltage	0.8756 [kV]
Sampling Interval	0.01
Initial Value of Vmpp	0.550 [kV]
Tracking Algorithm	Pertub & Observe
Name for Incremental Conductance	

Figure 11:MPPT Data

Since the inverter has 2 MPPT inputs, two strings with 22 modules in series are connected parallelly for each MPPT input.

For a single MPPT input:

$$\text{PV Array Short Circuit Current} = \text{PV Module } I_{SC} \times 2 = 9.81 \times 2 = \mathbf{19.62 \text{ A}}$$

$$\text{PV Array Open Circuit Voltage} = \text{PV Module } V_{OC} \times 22 = 39.8 \times 22 = \mathbf{875.6 \text{ V}}$$

MPP Tracking method used: **Perturb & Observe Method**

Household load is calculated using number of units consumed in past 10 months.

Average number of units consumed = 159.8 units

$$\text{Load} = \frac{159.8}{24 \times 30} \text{ kW} = 0.2219 \text{ kW}$$

3.1 Model Design Steps

The grid-connected PV system was simulated in PSCAD with a modular systematic method in order to capture the behaviour of the PV array, DC-DC boost converter, inverter and grid interface correctly. The PV array was initially set up according to the already designed string arrangement of 22 modules/array in series, and 2 strings/array in parallel, one MPPT channel. The parameters of the PV-array component used in PSCAD were configured with the number of series modules, the number of parallel strings, and per-module cell configuration. To initialize the model, standard reference conditions of irradiance of 1000 W/m² and cell temperature of 25°C were used.

Then computed the electrical properties of the integrated PV array are assigned with a sum of short-circuit current of **0.01962 kA** and an open-circuit voltage of **0.8756 kV**, as indicated in the Main Data window. These values are in line with what is expected of two series of 22 modules. The starting point of an MPP voltage was established at **0.550 kV** to help in convergence of the MPPT when the simulation is initiated. **Perturb and Observe (P&O)** algorithm has been chosen as the MPPT algorithm due to its simplicity and consistent tracking behavior with the irradiance changing slowly.

A DC-DC boost converter was added to connect the PV array to the inverter, and it was set to convert the PV voltage to the required level of DC-link under MPPT control. Two boost-converter stages and two identical PV array that were fed by two different irradiance and temperature inputs to each were modelled to represent the two MPPT inputs of the inverter. A three-phase voltage-source inverter was then attached to the DC-link in order to change the regulated DC power into AC power that may be injected into the grid.

Lastly, a circuit breaker with a timer, R-L load and connection to the grid were included to test islanded and grid-connected functionality. The PV array was driven with imported external CSV files of irradiance and temperature to enable the realistic evaluation of the system performance. On the whole, this type of modelling is a good representation of the electrical and control behaviour of a real grid-connected PV system.

3.2 Results Obtained

The irradiation and temperature profiles imported from the CSV file represent realistic time-varying environmental conditions for the PV array. The irradiation increases rapidly to a peak of around **1000–1100 W/m²** between few seconds, with noticeable fluctuations caused by intermittent shading or passing clouds. It then gradually decreases toward zero after some time. The module temperature follows a smoother trend, rising from **27°C** to about **31°C** during the high irradiance period and slowly decreasing as the irradiance drops. These variations provide a practical input for analyzing the dynamic behaviour of the PV system and the MPPT response under changing environmental conditions.

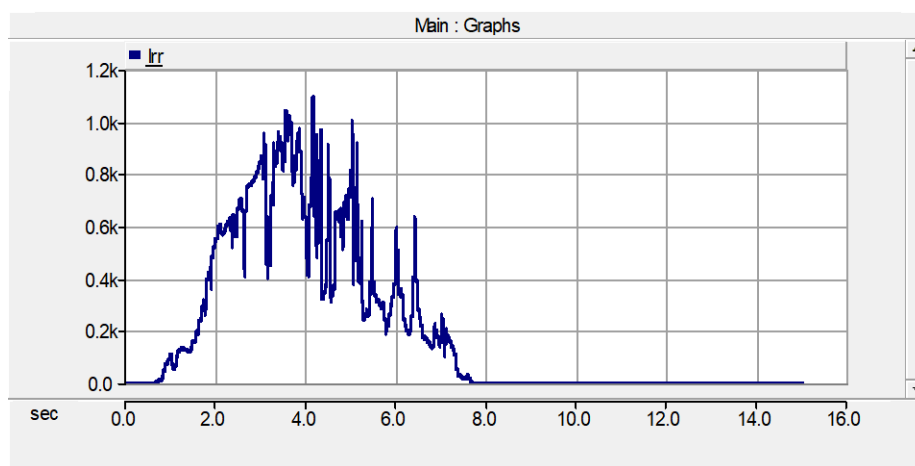


Figure 12:Solar Irradiation Variation

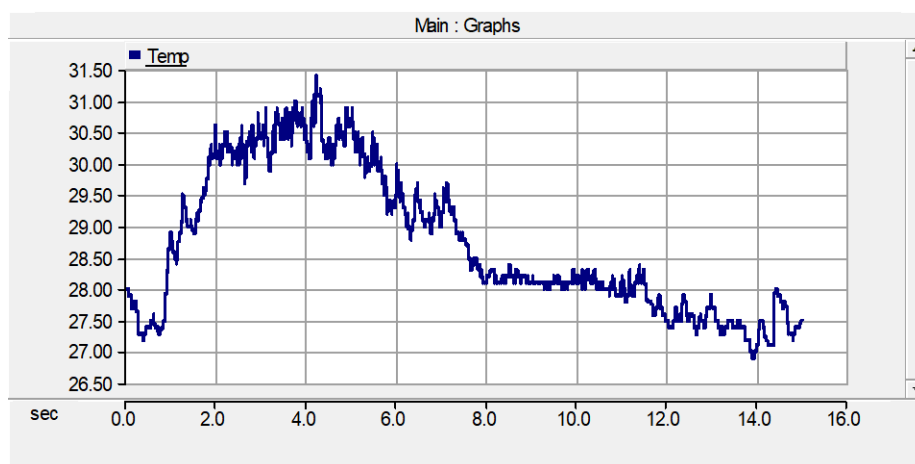


Figure 13:Temperature Variation

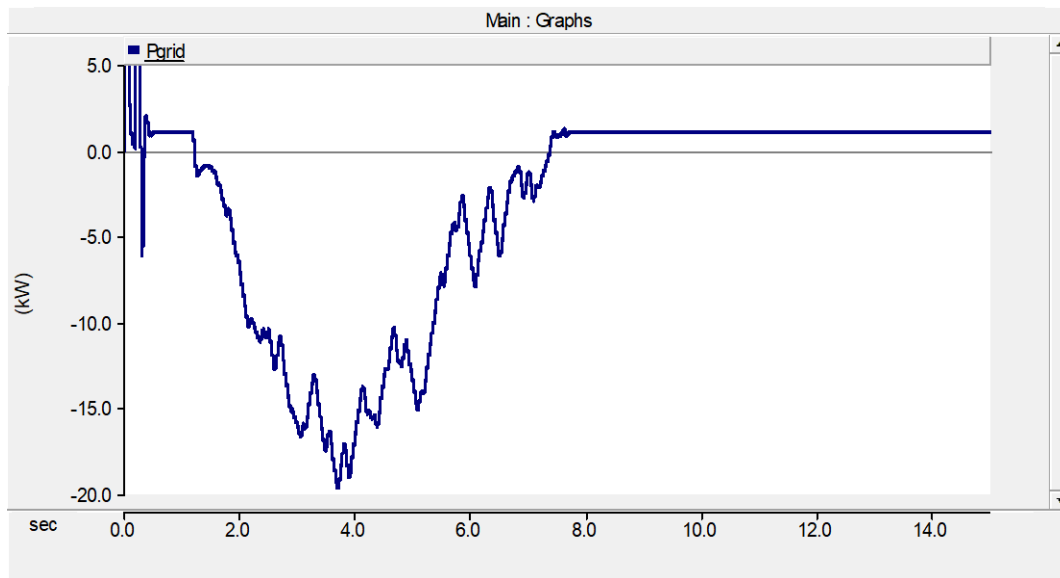


Figure 14:Grid Power Variation

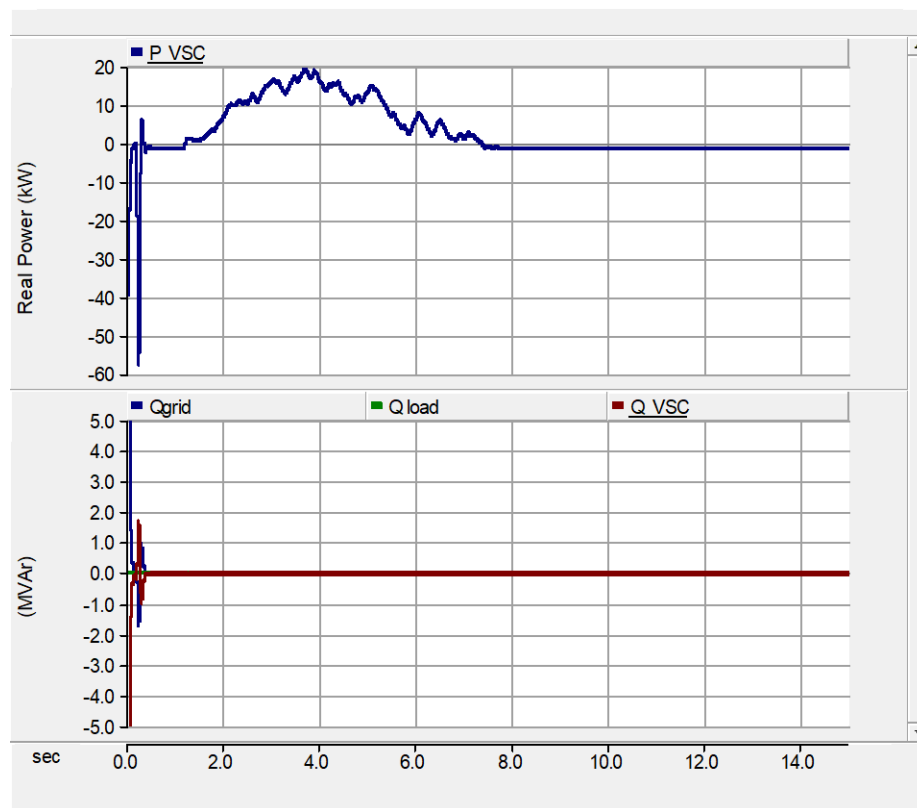


Figure 15:VSC & Grid Power (P&Q) Variation

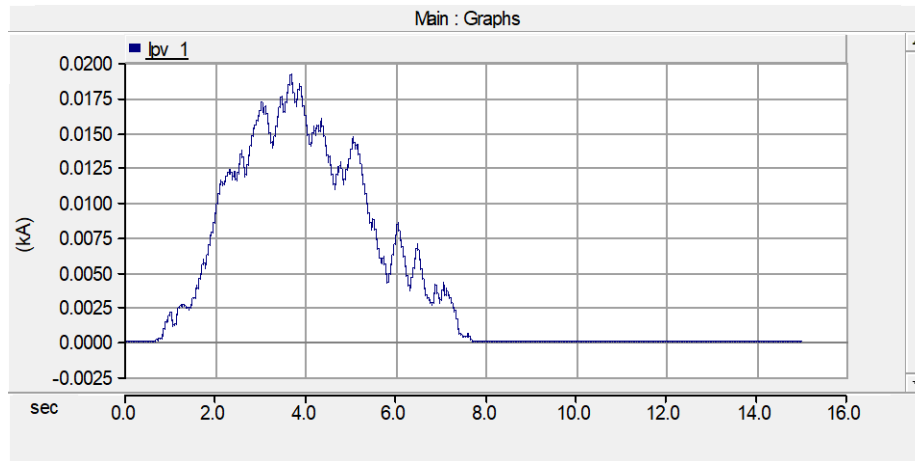


Figure 16:PV Array Current Variation

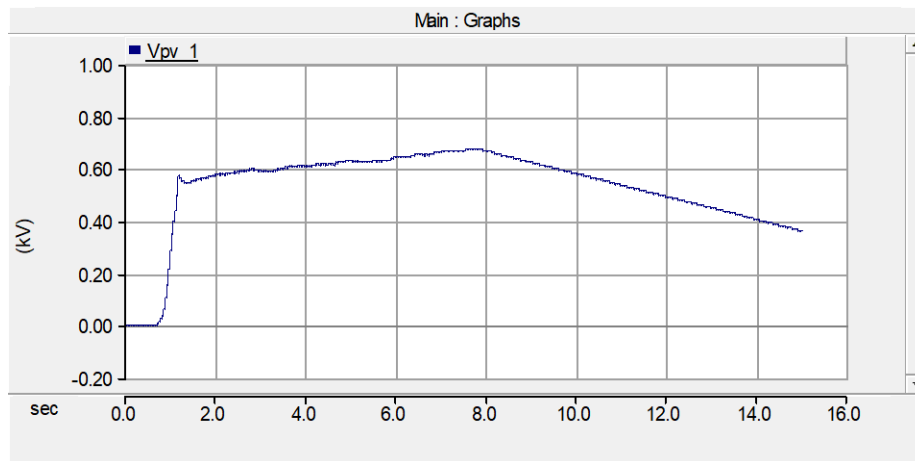


Figure 17:PV Array Voltage Variation

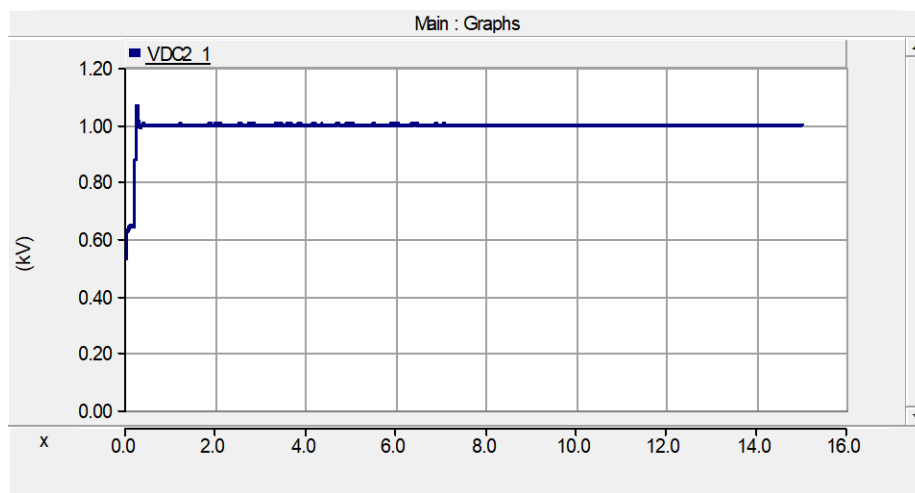


Figure 18:DC Link Voltage Variation

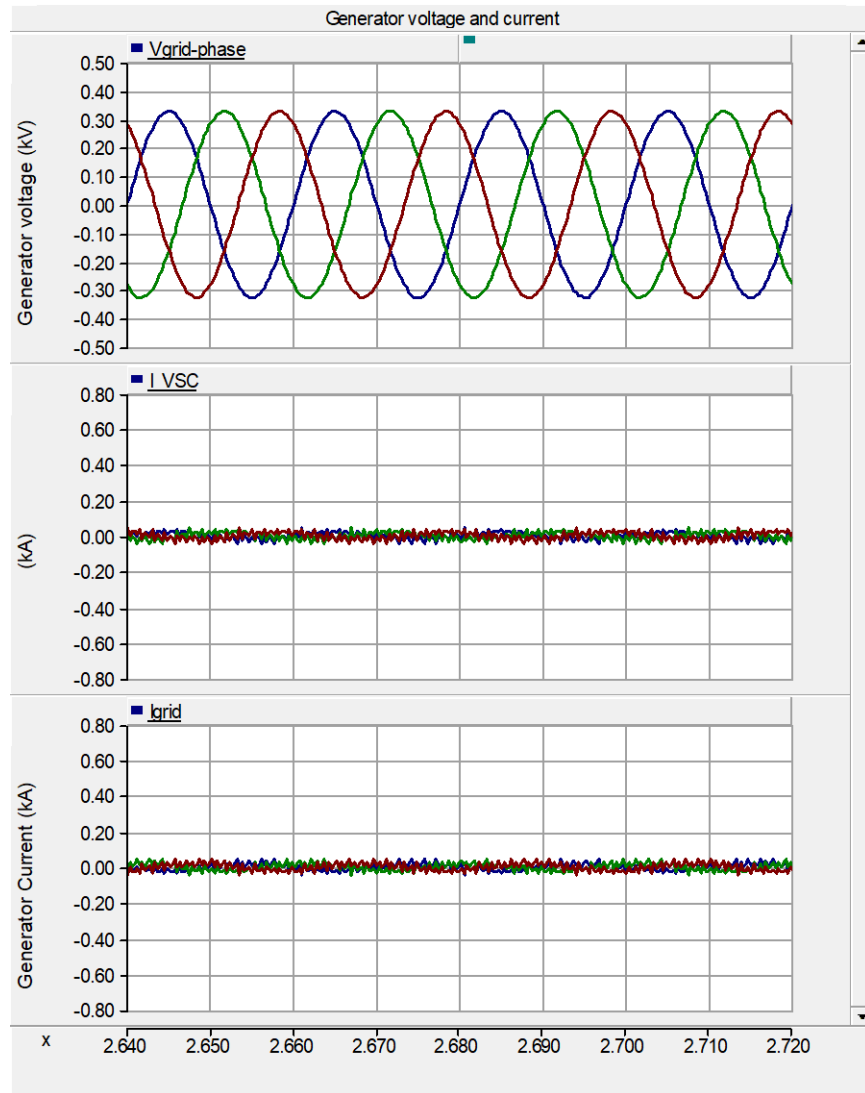


Figure 19:Grid Voltage, Inverter Voltage & Inverter Current Variation

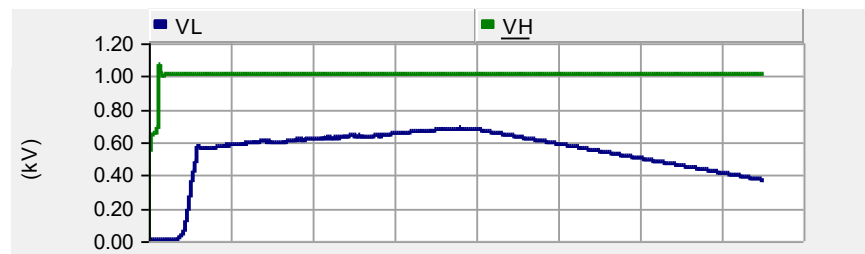


Figure 20:Boost Converter Input & Output Voltage Variation

3.3 Overall Discussion

The PSCAD simulation outcomes reveal varied dynamic behaviour of the PV system to different irradiance and temperature inputs based on CSV dataset. The current of PV array (**Figure 16**) soars up as the irradiance increases and peaks at the peak solar period. The PV array voltage (**Figure 17**) approaches the stabilization of the MPP voltage calculated by the MPPT controller and then slowly diminishes with a decrease in irradiance. In line with these findings, the DC-link voltage (**Figure 18**) is kept in a very tight control at about 1.0 - 1.1 kV which is a confirmation that the boost converter and inverter control loops are operational, and the DC-link is stable at the required level even at varying PV power levels.

Figure 20 of the boost converter voltage behaviour indicates the expected rise in voltages in the low input voltage area into the regulated DC-link level, then a slow decrease as the solar resource decreases. This has the effect of a seamless transition in power extraction that is regulated by the MPPT algorithm. **Figure 15** of inverter real and reactive power profile shows that the inverter injects real power, and reactive power is kept near zero (**unity power factor**). The actual power output is closely related to the availed PV power with the highest power at approximately **18 – 20 kW** and decreasing as the irradiance decreases.

Figure 14 Grid power curve indicates that a decrease in grid draw is experienced when high-irradiance periods occur and that a reversal is finally experienced when the PV generation surpasses the connected load. Bidirectional power flow behaviour is valid as the low irradiance once again causes an increase in grid import. The voltage and current waveforms (**Figure 19**) during the three phases are proof that the inverter delivers balanced sinusoidal currents in step with the grid voltage to guarantee the correct integration with the grid without distortion.

Altogether, the outcomes of the simulations confirm the fact that the PV array, the MPPT controller, the boost converter, the grid-tied inverter can be used in harmony. The system will react efficiently to environmental changes, ensure continuity of voltage, supply of maximum available power to the grid and have a good power quality of operation. It is verifying the effective work of the designed PV system in the PSCAD environment.

4 Annexes

4.1 Solar PV Module Data Sheet

Mono Multi Solutions

THE
DUOMAX^M PLUS⁺
DUAL GLASS 60-CELL MODULE

60 CELL
MONOCRYSTALLINE MODULE

280-315W
POWER OUTPUT RANGE

19.2%
MAXIMUM EFFICIENCY

0~+5W
POSITIVE POWER TOLERANCE

Founded in 1997, Trina Solar is the world's leading comprehensive solutions provider for solar energy. We believe close cooperation with our partners is critical to success. Trina Solar now distributes its PV products to over 60 countries all over the world. Trina is able to provide exceptional service to each customer in each market and supplement our innovative, reliable products with the backing of Trina as a strong, bankable partner. We are committed to building strategic, mutually beneficial collaboration with installers, developers, distributors and other partners.

Comprehensive Products And System Certificates

IEC61215/IEC61730/UL1703/IEC61701/IEC62716
ISO 9001: Quality Management System
ISO 14001: Environmental Management System
ISO14064: Greenhouse gases Emissions Verification
OHSAS18001: Occupation Health and Safety Management System



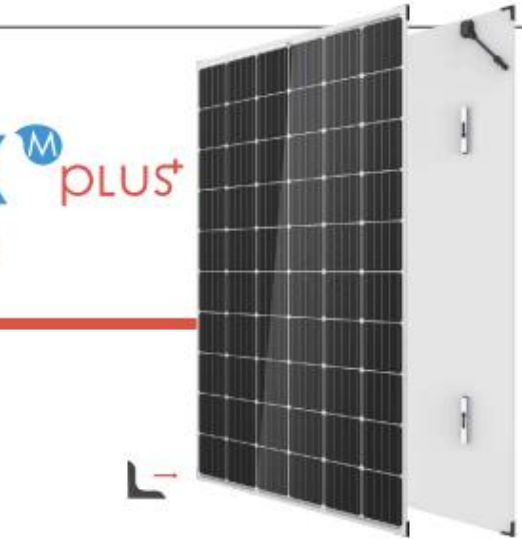








Trina solar





Highly reliable due to stringent quality control

- PID resistant and free of snail trails
- Increased module robustness to minimize micro-cracks
- 100% EL double inspection



Enhanced safety

- Fire class A certified by TUV Rheinland according to fire test IEC 61730-2/MST 23
- Certified for fire type I3 (UL 1703)



Increased value

- Higher maximum system voltage reduces BOS costs
- 30 year linear warranty
- 0.5% annual degradation
- Low thermal coefficients for more energy production at higher temperatures



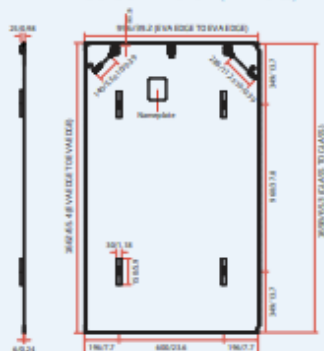
Certified to withstand the most challenging environmental conditions

- Module coating resistant to sand, acid, and alkali
- 2400 Pa wind load
- 5400 Pa snow load
- 35 mm hail stones at 97 km/h

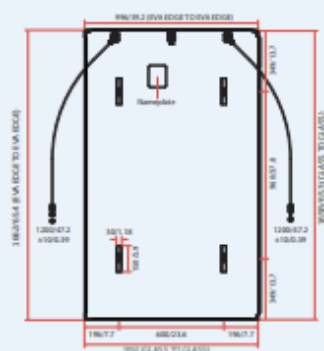


PRODUCTS	INSTALLATION METHOD	POWER RANGE
TSM-DEG5(I)	Clamp	280-315W
TSM-DEG5.40(II)	Gecko Grip	280-315W
TSM-DEG5.07(II)	Clamp	280-310W
TSM-DEG5.47(II)	Gecko Grip	280-310W

DIMENSIONS OF PV MODULE (mm/inches)

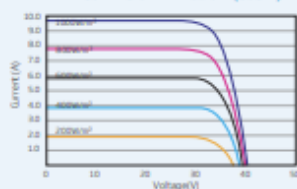


Back View (Portrait)

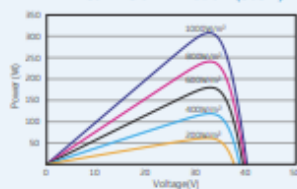


Back View (Landscape)

I-V CURVES OF PV MODULE(305W)



P-V CURVES OF PV MODULE(305W)



ELECTRICAL DATA (STC)

Peak Power Watts- P_{max} (Wp)*	280	285	290	295	300	305	310	315
Power Output Tolerance- P_{max} (W)				0	+5			
Maximum Power Voltage- V_{mp} (V)	31.7	31.8	32.2	32.5	32.7	32.9	33.1	33.3
Maximum Power Current- I_{mp} (A)	8.84	8.96	9.02	9.08	9.18	9.26	9.37	9.46
Open Circuit Voltage- V_{oc} (V)	38.4	38.5	38.7	39.4	39.8	40.0	40.2	40.4
Short Circuit Current- I_{sc} (A)	9.44	9.49	9.59	9.75	9.81	9.84	9.86	9.88
Module Efficiency η_m (%)	17.0	17.3	17.6	17.9	18.2	18.5	18.8	19.2

STC: Irradiance 1000W/m², Cell Temperature 25°C, Air Mass AM1.5
*Measuring tolerance: $\pm 3\%$

ELECTRICAL DATA (NOCT)

Maximum Power- P_{max} (Wp)	209	212	216	220	223	227	231	235
Maximum Power Voltage- V_{mp} (V)	29.4	29.5	29.9	30.1	30.3	30.5	30.7	30.9
Maximum Power Current- I_{mp} (A)	7.10	7.20	7.25	7.30	7.38	7.44	7.53	7.60
Open Circuit Voltage- V_{oc} (V)	35.7	35.8	36.0	36.6	37.0	37.2	37.4	37.5
Short Circuit Current- I_{sc} (A)	7.62	7.66	7.74	7.87	7.92	7.94	7.96	7.98

NOCT: Irradiance at 800W/m², Ambient Temperature 20°C, Wind Speed 1m/s.

MECHANICAL DATA

Solar Cells	Monocrystalline 156.75 × 156.75 mm (6 inches)
Cell Orientation	60 cells (6 × 10)
Module Dimensions	1658 × 992 × 6 mm (65.3 × 39.1 × 0.236 inches) 1662 × 996 × 6 mm with edge banding (65.4 × 39.2 × 0.236 inches) 1664 × 998 × 7.6 mm with corner protector (65.5 × 39.3 × 0.299 inches)*
Weight	23.5 kg (51.8 lb)
Front Glass	2.5 mm (0.10 inches), High Transmission, AR Coated Heat Strengthened Glass
EVA	White [DEG5(II), DEG5.40(II)]; Transparent [DEG5.07(II), DEG5.47(II)]
Back Glass	2.5 mm (0.10 inches), Heat Strengthened Glass
Frame	Frameless
J-Box	IP 67 or IP 68 rated
Cables	Photovoltaic Technology Cable 4.0 mm ² (0.006 inches ²) Portrait: 140/285 mm (5.5/11.2 inches) Landscape: 1200/1200 mm (47.2/47.2 inches)
Connector	Trina TS4 (1500V)
Fire Type	Type 13

TEMPERATURE RATINGS

NOCT (Nominal Operating Cell Temperature)	44°C ($\pm 2^\circ\text{C}$)
Temperature Coefficient of P_{max}	-0.39%/°C
Temperature Coefficient of V_{oc}	-0.29%/°C
Temperature Coefficient of I_{sc}	0.05%/°C

MAXIMUM RATINGS

Operational Temperature	-40~+85°C
Maximum System Voltage	1500V DC (IEC) 1000V DC (UL)
Max Series Fuse Rating	15A (Power $\leq 290\text{W}$) 20A (Power $\geq 295\text{W}$)

(DO NOT connect Fuse in Combiner Box with two or more strings in parallel connection)

WARRANTY

10 year Product Workmanship Warranty
30 year Linear Power Warranty

(Please refer to product warranty for details)

PACKAGING CONFIGURATION

Modules per box: 33 pieces
Modules per 40' container: 858 pieces

MORE OPTIONS

More Options	☐ 2.0mm Glass: 19.7 kg (43.4 lb) [Only for DEG5.40(II)/DEG5.47(II)] ☐ Compact AR Coating ☐ POE (Polyolefin Elastomer) Adhesive
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*Not applicable to slide-in racking solution

4.2 Inverter Data Sheet

SUN2000-12/15/17/20/25KTL-M5 Smart PV Controller



Active Safety

AI Powered Arcing Protection



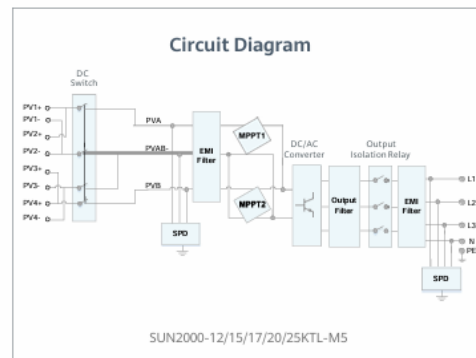
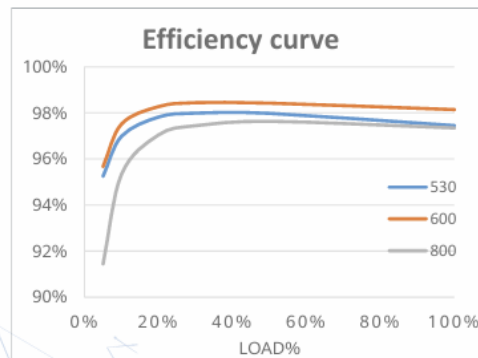
Higher Yields

Up to 30% More Energy with Optimizer



Flexible Communication

WLAN, Fast Ethernet, 4G
Communication Supported



SOLAR.HUAWEI.COM

Technical Specification

Technical Specification	SUN2000 -12KTL-M5	SUN2000 -15KTL-M5	SUN2000 -17KTL-M5	SUN2000 -20KTL-M5	SUN2000 -25KTL-M5
Efficiency					
Max. efficiency	98.4%	98.4%	98.4%	98.4%	98.4%
European weighted efficiency	97.9%	98.0%	98.1%	98.1%	98.2%
Input					
Recommended max. PV power ¹	18,000 Wp	22,500 Wp	25,500 Wp	30,000 Wp	37,500 Wp
Max. input voltage ²			1100 V		
Full-load MPPT voltage range	370V~800V	410V~800V	440V~800V	480V~800V	530~800V
MPPT Operating voltage range ³			200 V ~ 1000 V		
Start-up voltage			200 V		
Rated input voltage			600 V		
Max. input current per MPPT			30 A (two string) / 20 A (single string)		
Max. short-circuit current			40 A		
Number of MPP trackers			2		
Max. number of inputs			4		
Output					
Grid connection			Three phase		
Rated output power	12,000 W	15,000 W	17,000 W	20,000 W	25,000 W
Max. apparent power	13,200 W	16,500 VA	18,700 VA	22,000 VA	27,500 VA
Rated output voltage	220 Vac / 380 Vac, 230 Vac / 400 Vac, 239.6 Vac / 415Vac, 3W + N + PE				
Rated AC grid frequency	50 Hz / 60 Hz				
Max. output current	18.2A/380Vac	25.2A/380Vac	28.6A/380Vac	33.6A/380Vac	42.0A/380Vac
	17.3A/400Vac	23.9A/400Vac	27.1A/400Vac	31.9A/400Vac	39.9A/400Vac
	16.7A/415Vac	23.1A/415Vac	26.1A/415Vac	30.8A/415Vac	38.5A/415Vac
Adjustable power factor	0.8 leading ... 0.8 lagging				
Max. total harmonic distortion	≤ 3 %				

Features & Protections	
Overvoltage Category	PV II/AC III
Input-side disconnection device	Yes
Anti-islanding protection	Yes
AC over-current protection	Yes
DC reverse-polarity protection	Yes
String fault detection	Yes
DC surge protection	TYPE II
AC surge protection	CLASS II
Residual current monitoring unit	Yes
Arc fault protection	Yes
Ripple control ripple control	Yes
Integrated PID recovery ⁴	Yes

General Data	
Operation temperature range	-25 ~ + 60 °C (-13 °F ~ 140 °F)
Relative humidity	0 % RH ~ 100% RH
Max. operating altitude	0 ~ 4,000 m (13,123 ft.) (Derating above 2000 m)
Cooling	Smart air cooling
Display	LED Indicators; Integrated WLAN + FusionSolar App
Communication	RS485; WLAN/Ethernet via Smart Dongle-WLAN-FE (Optional) 4G / 3G / 2G via Smart Dongle-4G (Optional)
Weight (with mounting plate)	21kg (46.4 lb)
Dimensions (W x H x D) (incl. mounting plate)	546 x 460 x 228mm (21.5 x 18.1 x 9.0 inch)
Degree of protection	IP66

Optimizer Compatibility	
DC MBUS compatible optimizer	SUN2000-450W-P, SUN2000-450W-P2, SUN2000-600W-P, SUN2000-1300W-P, SUN2000-1100W-P

Standard Compliance (more available upon request)	
Safety	EN/IEC 62109-1, EN/IEC 62109-2
Grid connection standards	G99, EN 50549, CEI 0-21, CEI 0-16, VDE-AR-N-4105, VDE-AR-N-4110, C10/11, ABNT, VFR 2019, UNE 217001, UNE 217002, RD 244, TOR D4, IEC61727, IEC62116

¹ Inverter max input PV power is 40,000 Wp when long strings are designed and fully connected with SUN2000-450W-P power optimizers.

² The maximum input voltage is the upper limit of the DC voltage. Any higher input DC voltage would probably damage inverter.

³ Any DC input voltage beyond the operating voltage range may result in inverter improper operating.

⁴ SUN2000-12~20KTL-M2 raises potential between PV- and ground to above zero through integrated PID recovery function to recover module degradation from PID. Supported module types include: P-type (mono, poly)

Version (No. 01- (20)90716)

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